An Abstract Constraint Chart for Measurement-Limited State Representations

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Abstract:

Reasoning about complex systems is often hindered by the presence of incompatible constraints, non-commuting operations, and limited intuitive access to high-dimensional state descriptions. In quantum and other constraint-rich domains, these difficulties are exacerbated by the mismatch between formal mathematical representations and the operational distinctions that are actually accessible under restricted measurement or control capabilities. This work introduces the Abstract Constraint Chart (ACC), a representational tool inspired by the classical Smith chart, designed to organize and visualize equivalence classes of states induced by constraint-limited distinguishability. Rather than mapping individual states or defining new dynamics, the ACC represents coarse-grained structure arising from explicitly specified equivalence relations, such as measurement-indistinguishability under restricted admissible measurements. Transformations within the ACC correspond to admissible operations that induce well-defined transitions between equivalence classes. Critically, these encode accessibility and directional change without invoking time evolution, mechanism, or physical causality. The Framework preserves relational structure, such as partial ordering, adjacency, and boundary behaviour, while deliberately discarding metric, dynamical, and fine-grained state information. A minimal toy instantiation is presented for a single qubit under restricted measurement capabilities, with a tighter example using a 2D Ising lattice under local observables further illustrating constraint-imposed trade-offs. The ACC is proposed as a general representational aid for reasoning and comparison across constrained systems, not as a predictive model or physical theory.

Keywords: quantum information foundation; generalized probabilistic theories (GPTs); equivalence classes; order structure

Section 1. Introduction

1.1 Motivation

In many areas of physics, mathematics, and information science, the practical difficulty of reasoning about systems does not arise from a lack of formal description, but from an excess of structure relative to what is operationally accessible. While a system may admit a complete mathematical representation, the distinctions that can actually be drawn are often constrained by limited measurements, incompatible observables, or restricted classes of admissible operations. As a result, full state-space descriptions may obscure rather than clarify the structure that governs meaningful comparison and reasoning.

Quantum mechanics provides a canonical example. Although density operators or state vectors furnish mathematically exhaustive representations, experimental and conceptual access is mediated by restricted measurement sets and non-commuting observables. In such contexts, many formally distinct states are operationally indistinguishable, while others differ only in ways that cannot be simultaneously resolved. Reasoning at the level of individual states therefore risks importing distinctions that are not supported by the available constraints.

Similar issues arise outside quantum theory. In control systems with limited actuation, in communication protocols with restricted observables, and in constrained optimization problems where only partial orderings are relevant, the effective structure is determined less by the full configuration space than by the relations that survive under explicit constraints. In these settings, representations that preserve only relational features, such as order, adjacency, and boundary structure, can be more informative than metric or dynamical descriptions.

Historically, a number of successful representational tools have addressed this tension by organizing equivalence classes rather than individual configurations. The Smith chart is a well-known example: it visualizes impedance relationships by collapsing physically distinct configurations into equivalence classes defined by measurement and transformation constraints, while preserving only the structure relevant for reasoning and design. Crucially, such charts do not model dynamics or mechanisms; they function as disciplined aids for navigating constraint-defined structure.[1]

The motivation for the present work is to articulate a similarly disciplined representational object for systems governed by explicit measurement and transformation constraints, where meaningful distinctions arise at the level of equivalence classes rather than individual states.

1.2 Scope and Intent

The Abstract Constraint Chart (ACC) is introduced as a representational framework, not as a physical theory, dynamical model, or predictive formalism. Its purpose is to organize and make explicit the relational structure that arises when a state space is viewed through a specified set of admissible measurements and operations. The ACC does not posit new principles, modify existing domain theories, or offer explanatory mechanisms. All empirical, causal, or dynamical content remains entirely within the underlying theory appropriate to the system under consideration.

At the core of the ACC is a deliberately modest move: states that are operationally indistinguishable under a given constraint regime are identified via an explicit equivalence relation, and reasoning is carried out on the resulting quotient structure. This approach reflects a minimal operational insight, that distinctions without measurement support should not appear in the representation, without committing to any broader interpretive stance [8] [11]. The resulting chart encodes only those relations that are invariant under the imposed constraints.

The ACC is intentionally agnostic with respect to ontology, dynamics, and geometry. It does not assert that equivalence classes correspond to physically identical states, nor that preserved orderings reflect causal or temporal structure. It does not introduce metrics, probabilities, or trajectories, and it makes no claim of completeness or optimality. Its role is purely organizational: to clarify which distinctions survive under constraint and how those distinctions relate to one another.

This restraint is deliberate. Representational frameworks often precede, rather than replace, theory development, providing shared structure without predictive ambition [S1, S2]. In this sense, the ACC is closer in spirit to chart-based or order-theoretic representations than to models intended for explanation or inference. Its value lies in making constraint-induced structure explicit and communicable, not in deriving new empirical consequences.

While the ACC is illustrated in this paper using minimal examples, it is not inherently quantum or tied to any specific domain. The formal construction applies equally to classical, probabilistic, or abstract state spaces, provided that admissible measurements and transformations can be specified. Any domain-specific interpretation must therefore be supplied externally, and no claim is made that the ACC selects or privileges particular constraint regimes.

1.3 Structure of the Paper

The remainder of this paper develops the Abstract Constraint Chart in a deliberately incremental manner.

Section 2 introduces the formal setting. It specifies the objects of representation, defines equivalence relations induced by admissible measurement sets, and establishes the quotient space on which the ACC is defined. Emphasis is placed on generality and on the explicit dependence of all structure on stated constraints.

Section 3 identifies the minimal relational structures preserved by the ACC. These include measurement-induced orderings, adjacency relations corresponding to local distinguishability, boundary elements arising from constraint-induced limits, and monotonic relations under admissible transformations. Structures that are intentionally excluded, such as metrics, dynamics, and causal interpretation, are stated explicitly.

Section 4 defines admissible transformations and their induced action on equivalence classes. The distinction between accessibility relations and physical dynamics is emphasized to prevent misinterpretation of chart transitions.

Section 5 presents a toy instantiations, beginning with a 2D Ising lattice under restricted local measurements, to illustrate the representational construction in a familiar constrained setting.

Section 6 delineates limitations, boundary conditions, and domains of applicability, clarifying when the ACC is informative and when it becomes trivial or inappropriate.

Section 7 concludes with a brief summary and outlook, situating the ACC as a disciplined reasoning aid for constraint-rich systems.

Appendix A collects formal propositions and remarks establishing well-definedness, order-theoretic properties, boundary structure, monotonicity without dynamics, and conditions under which the ACC trivializes. The appendix introduces no new assumptions and may be omitted by readers interested only in conceptual aspects.

Section 2. Formal Setting

This section specifies the formal objects underlying the Abstract Constraint Chart (ACC). The construction is intentionally minimal: it introduces only those structures required to define operational equivalence, quotienting, and the relational features preserved by the chart. No assumptions are made regarding geometry, dynamics, or ontology beyond what is explicitly stated.

2.1 Objects of Representation

Let $\mathcal{S}$ denote the set of admissible states of a system. The ACC places no restrictions on the internal structure of $\mathcal{S}$ beyond the requirement that elements of $\mathcal{S}$ admit evaluation by a specified class of admissible measurements. In particular, $\mathcal{S}$ need not carry a metric, topology, or dynamical law.

In the illustrative examples presented later, $\mathcal{S}$ may be taken to be configurations of a lattice or density operators acting on a finite-dimensional Hilbert space, following standard operational treatments. [10]

[12] This choice is purely illustrative. The formal construction applies equally to classical state spaces, generalized probabilistic theories, or abstract collections of configurations, provided that states are identified only through admissible measurement statistics.

Crucially, the ACC does not represent individual elements of $\mathcal{S}$ directly. Its representational objects are equivalence classes of states induced by explicitly stated constraints. Any structure present in the ACC must therefore descend from relations that are invariant across all representatives of a class. This design ensures that the chart reflects only distinctions that are operationally meaningful under the imposed constraints, and no others.

2.2 Equivalence Relations Induced by Constraints

Let $\mathcal{M}_{allowed}$ denote a specified set of admissible measurements or evaluative functionals on $\mathcal{S}$. These measurements encode the operational constraints under which states may be distinguished. No assumption is made that $\mathcal{M}_{allowed}$ is informationally complete, closed under algebraic operations, or jointly compatible.

An equivalence relation $\sim$ on $\mathcal{S}$ is defined by measurement indistinguishability with respect to $\mathcal{M}_{allowed}$. For ρ_1 , $\rho_2 \in \mathcal{S}$,

$$\rho_1 \sim \rho_2 \iff \mathrm{Tr}(M\rho_1) = \mathrm{Tr}(M\rho_2) \quad \forall M \in \mathcal{M}_{\text{allowed}} .$$

This definition follows standard operational logic [12]: states are identified if and only if they agree on all accessible measurement statistics.

The equivalence classes induced by $\sim$ are denoted $[\rho]$, and the quotient set $\mathcal{S}/\sim$ constitutes the representational domain of the ACC. The identification effected by $\sim$ is strictly operational. It does not assert physical identity, ontological equivalence, or redundancy of the underlying states; it records only the resolution limits imposed by the chosen measurement set.

Different choices of $\mathcal{M}_{allowed}$ generally induce different equivalence relations and therefore different quotient structures. All ACC representations are thus conditional on explicit constraint specification, and no canonical or observer-independent chart is implied.

2.3 Well-Definedness on the Quotient

The use of equivalence classes requires that all representational features of the ACC be well-defined on $\mathcal{S}/\sim$, independent of the choice of representative. This condition is satisfied by construction.

Because expectation values of admissible measurements are invariant across each equivalence class, any relation defined purely in terms of such expectation values descends unambiguously to the quotient.[16][18] This includes ordering relations, boundary identification, and monotonic relations discussed in subsequent sections.

Similarly, admissible transformations are required to respect the equivalence relation: if

$\rho_1 \sim \rho_2$, then any admissible transformation must map them to equivalent outputs. This consistency condition ensures that transformations induce well-defined mappings on
$\mathcal{S}/\sim$ rather than representation-dependent relations. The ACC encodes only these induced class-level mappings.

2.4 Generality and Domain Independence

The formal setting of the ACC is deliberately domain-agnostic. The construction relies only on three ingredients: a state set $\mathcal{S}$, an admissible measurement set $\mathcal{M}_{allowed}$, and an induced equivalence relation defined by indistinguishability. No appeal is made to specifically quantum notions such as superposition, entanglement, or wave-function collapse, nor to classical phase-space structure.
This generality aligns the ACC with established mathematical practices of quotienting and order-based reasoning under partial information.[15][16] It also ensures that any additional interpretation, physical, informational, or experimental, must be supplied externally by the relevant domain theory.

2.5 Dependence on Explicit Constraint Specification

All structure in the ACC depends explicitly on the chosen constraints. Refining, enlarging, or restricting $\mathcal{M}_{allowed}$ alters the equivalence relation and therefore the representational content of the chart. In the limiting cases, informationally complete measurement sets collapse the quotient to the original state space, while degenerate measurement sets collapse all states into a single class.
This dependence is not a defect but a defining feature. The ACC is designed to make the consequences of constraint choice explicit, allowing comparison across regimes without importing unstated assumptions.

Section 3. Preserved Structure in the Abstract Constraint Chart

The Abstract Constraint Chart (ACC) represents equivalence classes of states induced by constraint-limited distinguishability. As a quotient representation, it does not inherit the full structure of the underlying state space $\mathcal{S}$. Instead, it preserves a deliberately restricted set of relational features that remain well-defined on the quotient $\mathcal{S}/\sim$. This section specifies those preserved structures and states explicitly what is excluded.

3.1 Partial Ordering Induced by Admissible Measurements

Each admissible measurement $M \in \mathcal{M}_{allowed}$ induces an order relation on equivalence classes via expectation values. For classes ρ_a and ρ_b , define $[\rho_a] \preceq_M [\rho_b] \iff \mathrm{Tr}(M\rho_a) \leq \mathrm{Tr}(M\rho_b)$.
By construction, this relation is invariant under choice of representative and therefore well-defined on the quotient.
In general, $\preceq M$ defines a preorder rather than a total or antisymmetric order. Distinct equivalence classes may yield identical expectation values and thus be mutually comparable in both directions. The ACC preserves this preorder structure without refinement unless explicitly specified. Order-theoretic

language is used intentionally: the chart records relative positioning under admissible measurements without introducing metric separation or scalar aggregation.[15][17]

Different admissible measurements typically induce incompatible preorders. The ACC does not attempt to reconcile or unify these orderings into a single global structure. Their coexistence reflects the constraint regime itself and mirrors familiar incompatibilities in operational settings.[12]

3.2 Adjacency and Local Distinguishability

Beyond ordering, the ACC preserves adjacency relations that encode local distinguishability under constraints. Two equivalence classes are adjacent if there exists an admissible transformation that effects an arbitrarily small change in the expectation value of at least one admissible measurement while remaining within the constraint regime.

Adjacency is thus defined operationally, not metrically. It captures which distinctions can be accessed incrementally and which require discontinuous or forbidden transitions. The ACC records adjacency as a primitive relational feature, without assigning distances, angles, or neighbourhoods in a topological sense.[16][15]

This notion is particularly relevant in constraint-rich systems, where small changes in accessible statistics may correspond to large or indeterminate changes in the underlying state space. By preserving adjacency rather than metric proximity, the ACC avoids importing geometric assumptions that are not supported by the constraints.

3.3 Boundary Structure and Constraint-Induced Limits

Admissible measurements often have bounded expectation values on $\mathscr{S}$. Equivalence classes that saturate these bounds define boundary elements of the ACC with respect to the corresponding measurement-induced preorder.

These boundaries are preserved as extremal positions within the ordering, not as geometric edges or physically distinguished states. Their significance is entirely constraint-induced: they mark limits of distinguishability under the specified measurement set and may shift or disappear if constraints are modified.[15][26]

The ACC preserves the existence and relative placement of such boundary classes. No claim is made that boundary elements correspond to pure states, eigenstates, or otherwise privileged configurations in the underlying theory. The term “boundary” is used strictly in an order-theoretic sense.

3.4 Monotonicity Under Admissible Transformations

Although the ACC does not encode dynamics, it preserves monotonic relations induced by admissible transformations. If a transformation consistently increases or decreases the expectation value of an admissible measurement across an equivalence class, the corresponding directional relation between classes is retained.

Formally, a transformation that is monotone with respect to a measurement M induces an order-preserving map on the quotient. The ACC records only the direction and consistency of such changes. It does not encode rates, generators, composition laws, or temporal ordering.[17][19]
This monotonic structure supports reasoning about accessibility, what can be reached from what under constraints, without introducing causal or dynamical interpretation. Directionality here is relational, not temporal.

3.5 Explicitly Unpreserved Structure

For clarity, the ACC does not preserve: metric distances or angles, inner products or geometric embeddings, probabilistic weights beyond order comparisons, dynamical trajectories or time parameters, causal structure or inference relations.
These exclusions are intentional and constitutive of the framework. Any attempt to reintroduce such structure would require additional assumptions beyond the specified constraints and would therefore fall outside the ACC's representational scope. In particular, no causal interpretation is implied or supported by chart relations.[S3]

3.6 Preserved vs. Discarded Structure (Summary)

For emphasis, the ACC preserves: equivalence classes under explicit constraints, measurement-induced preorders, adjacency relations of local distinguishability, boundary elements as extremal order positions, monotonic accessibility relations.
It discards: geometry, metrics, and topology, dynamics, rates, and trajectories, causal and ontological interpretation.
This separation is not a limitation to be remedied but a design constraint. By preserving only order-theoretic and relational structure, the ACC makes explicit exactly what survives constraint-induced reduction and nothing more.

Section 4. Admissible Transformations and Class Transitions

The Abstract Constraint Chart (ACC) represents how equivalence classes relate under transformations that respect the imposed measurement constraints. This section formalizes admissible transformations, specifies the induced mappings on the quotient space, and clarifies the limited sense in which "transitions" are represented. No dynamical, causal, or temporal interpretation is introduced.

4.1 Admissible Transformations

An admissible transformation is defined as any mapping $T : \mathcal{S} \to \mathcal{S}$ that preserves the equivalence relation induced by $\mathcal{M}_{allowed}$. Explicitly, T is admissible if $\rho_1 \sim \rho_2 \implies T(\rho_1) \sim (\rho_2)$

This condition ensures that the action of T is well-defined on equivalence classes and therefore induces a mapping $\widetilde{T} : \mathcal{S}/\sim \,\to \mathcal{S}/\sim$. The ACC encodes only the induced mapping $\widetilde{T}$, not the underlying action of T on representatives. Any representational content that depends on representative choice is excluded by design.

No assumption is made that admissible transformations form a group, semigroup, or category. Composition, invertibility, and continuity are not required. The admissibility criterion is purely operational and constraint-relative.

4.2 Class Transitions Without Dynamics

A class transition in the ACC is the relation $[\rho] \mapsto [\rho']$ induced by an admissible transformation. Such transitions encode the possibility of moving between equivalence classes under the specified constraints. They do not encode time evolution, process duration, or causal mechanism.

The term "transition" is used descriptively rather than dynamically. It denotes a relation between equivalence classes under an allowed mapping, not a physical process unfolding in time. Multiple admissible transformations may induce identical class-level transitions, and the ACC does not distinguish between them [17][19].

This abstraction allows the ACC to represent accessibility relations without importing assumptions about underlying dynamics. In particular, it avoids conflating monotonic change with temporal flow, a distinction emphasized throughout the framework.

4.3 Monotonicity and Order Preservation

Certain admissible transformations preserve or reverse the preorders induced by admissible measurements. If a transformation satisfies $[\rho_a] \preceq_M [\rho_b] \implies T([\rho_a]) \preceq_M T([\rho_b])$, it is order-preserving with respect to measurement M. The ACC records this monotonicity as a relational feature between classes.

Only the existence and direction of monotonicity are represented. The ACC does not encode the magnitude of change, rates, or whether the transformation is generated continuously or discretely. These omissions are intentional and aligned with the framework's representational scope.

Transformations that violate monotonicity for a given measurement are not excluded but do not contribute order-preserving structure for that measurement. The ACC remains agnostic regarding their detailed behaviour.

4.4 Fixed Classes and Invariant Regions

Some equivalence classes may be invariant under a given admissible transformation, satisfying $T([\rho]) = [\rho]$. Such fixed classes are represented as self-maps in the ACC. Their presence indicates constraint-induced stability, not equilibrium or physical conservation. No dynamical interpretation is implied.

More generally, subsets of equivalence classes may be invariant as a set, forming regions of the chart closed under admissible transformations. The ACC records these invariant regions as relational features without assigning additional structure or interpretation [18].

4.5 Scope and Exclusions

The ACC representation of transformations is intentionally limited. It does not preserve: temporal ordering or time parameters, causal relations or intervention semantics, generators, Hamiltonians, or stochastic kernels, compositional laws beyond induced class mappings.

Any interpretation of class transitions as physical evolution must be supplied externally and justified independently. Within the ACC, transformations are representational devices for relating equivalence classes under explicit constraints, nothing more.

Section 5. A Toy Instantiation: 2D Ising Lattice Under Restricted Local Measurements

This section presents illustrative instantiations of the Abstract Constraint Chart (ACC) using a 2D Ising lattice under restricted measurement access. The purpose is not to motivate the framework or to derive physical insight, but to demonstrate how the formal construction operates in a familiar constrained setting. The ACC remains fully defined independently of these examples.

5.1 State Space and Measurement Constraints

Consider a finite 2D Ising lattice with configurations $\sigma \in \{\pm 1\}^N$ (N sites on a toroidal grid for periodic boundary conditions). While this state space admits exact solutions in many regimes, no such global structure is used in the ACC construction and is treated as inessential for the purposes of representation [10][12].

Suppose that the admissible measurement set $\mathscr{M}_{allowed}$ consists of local plaquette energy observables $E_p = -\sigma_i\sigma_j - \sigma_k\sigma_l$ for each 2×2 plaquette p and/or staggered magnetization on one sub-lattice $m_s = \frac{1}{2N}\sum_{i\in s}\sigma_i$. Operationally, this restriction reflects a constraint under which only these local expectation values are accessible[11][13].

This constraint defines the regime under which states may be distinguished and therefore determines the induced equivalence relation.

5.2 Induced Equivalence Classes

Under the specified measurement constraints, two configurations ρ_1 and ρ_2 are equivalent if and only if they yield the same expectation values for all observables in $\mathscr{M}_{allowed}$. All configurations sharing common values of the allowed local expectations are therefore identified into a single equivalence class. The resulting quotient $\mathscr{S}/\sim$ organizes the full lattice configuration space into equivalence classes indexed by the vector of accessible expectation values. This identification discards distinctions associated with measurements outside $\mathscr{M}_{allowed}$, consistent with the operational constraint [11][12].
No claim is made that the equivalence classes correspond to physically homogeneous or ontologically meaningful subsets of states. They are defined solely by indistinguishability under the allowed measurements.

5.3 Preserved Ordering and Boundary Structure

The admissible measurements induce natural preorders on the equivalence classes via ordering of expectation values. Classes may be ordered by increasing local plaquette energies or staggered magnetization, yielding (partial) preorders on the quotient.
The extremal equivalence classes correspond to saturation of the measurement bounds (e.g., all plaquettes fully aligned or maximally frustrated). These classes form boundary elements of the ACC with respect to the induced ordering. Their boundary status reflects only the limits of the admissible measurements and does not imply global extremality or any privileged physical role [15] [26].
Adjacency relations correspond to infinitesimal changes in the accessible expectation values (via local spin flips or cluster updates) and are preserved without invoking any metric or geometric structure [17].

5.4 Admissible Transformations and Class Transitions

Admissible transformations in this setting are those mappings on the lattice configurations that preserve equivalence under the restricted measurements. Any transformation that leaves the allowed local expectation values invariant induces the identity mapping on the ACC, while transformations that monotonically increase or decrease those expectations induce order-preserving class transitions.
The ACC records only the induced class-level relations. It does not distinguish between different operations that produce the same change in expectation value, nor does it encode temporal ordering, rates, or generators [17][19].
Transformations that alter inaccessible degrees of freedom without affecting the allowed expectations are invisible to the chart.

5.5 Interpretation of the Chart Representation

The resulting ACC representation for this toy instantiation is a collection of equivalence classes equipped with measurement-induced preorders, adjacency relations, boundary elements, and monotonic accessibility maps. This structure reflects only the imposed measurement constraints and should not be interpreted as a reduced dynamical model, effective theory, or physical explanation.

The example serves solely to illustrate how constraint-induced equivalence, ordering, adjacency, and boundary structure appear in a concrete lattice setting.

Section 6. Limitations, Boundary Conditions, and Scope of Applicability

The Abstract Constraint Chart (ACC) is deliberately limited in scope. Its purpose is not to provide a complete or optimal representation of physical systems, but to formalize how explicit measurement constraints induce equivalence classes and preserve a restricted set of relational structures. This section clarifies the boundaries of applicability of the framework and the conditions under which it is not intended to be used.

6.1 Dependence on Explicit Constraints

All structure represented in an ACC depends explicitly on the choice of admissible measurements. Refining, enlarging, or restricting the measurement set alters the induced equivalence relation and therefore the resulting chart. No ACC representation is invariant under arbitrary changes of constraint, nor is any single chart privileged as canonical [15][16].

This dependence is a constitutive feature of the framework. The ACC is designed to make constraint-induced information loss explicit, not to minimize or eliminate it. Comparisons between charts constructed under different constraint regimes are meaningful only when the differing constraints are themselves part of the analysis.

6.2 Inapplicability to Fully Accessible or Metric-Dominated Systems

In regimes where measurement access is informationally complete, the induced equivalence relation collapses to identity and the ACC reduces trivially to the original state space. In such cases, the framework provides no representational advantage and is not intended to replace existing geometric, probabilistic, or dynamical formalisms [10][12].

Similarly, in systems where metric structure, continuity, or quantitative distance measures are essential to the analysis, the ACC may discard information that is operationally or theoretically central. The framework intentionally avoids encoding metric or topological structure and should not be applied where such structure is required for correct reasoning.

6.3 No Claims of Completeness or Optimality

The ACC makes no claims of representational completeness, optimality, or minimality in an information-theoretic sense. It does not assert that preserved order, adjacency, and boundary relations exhaust all meaningful content under constraint, nor that discarded structure is irrelevant in other contexts [17][18]. The framework is therefore not proposed as a universal reduction scheme or as a replacement for domain-specific models. It is a disciplined representational tool whose utility depends on the questions being asked and the constraints being imposed.

6.4 Domain-Specific Interpretation Required

Interpretation of any ACC representation must be supplied externally by the relevant domain theory. The chart itself encodes no physical ontology, causal structure, or experimental narrative. Relations depicted in the ACC should not be read as physical processes, mechanisms, or explanations without additional justification [19][26].

In particular, monotonic relations and class transitions do not imply temporal evolution or causal influence. Any such interpretation lies outside the scope of the framework and must be argued independently.

6.5 Summary of Applicability

The ACC is applicable when: measurement access is explicitly constrained, equivalence under indistinguishability is central, relational structure is sufficient for the intended analysis.

It is not applicable when: full state access is available, metric or dynamical detail is essential, interpretive or explanatory claims are required.

These limitations are not shortcomings but design constraints. They define the narrow representational role the ACC is intended to play.

Section 7. Conclusion and Outlook

This work has introduced the Abstract Constraint Chart (ACC) as a representational framework for organizing state descriptions under explicit measurement constraints. By quotienting a state space according to operational indistinguishability and preserving only order, adjacency, and boundary relations, the ACC makes precise what structural information survives constraint-induced reduction.

The framework is intentionally limited. It does not model dynamics, encode geometry, or support causal interpretation. Instead, it provides a disciplined means of reasoning about equivalence classes and relational structure when access to a system is restricted. In this respect, the ACC aligns with established operational and order-theoretic approaches while remaining agnostic about the underlying physical theory [10][12][15].

The minimal examples, including a tighter 2D Ising lattice instantiation under restricted local measurements, illustrate how familiar systems may be represented within the ACC without invoking domain-specific structure beyond the explicitly chosen constraints. The examples are not evidentiary and do not motivate the framework; they serve only to demonstrate internal consistency and practical construction under simple constraint regimes.

Future work may apply the ACC to other constrained settings, including higher-dimensional lattice models, coarse-grained classical systems, or abstract operational theories. Such applications would use the ACC as a representational tool rather than extending or redefining the framework itself. Any additional interpretation or explanatory content would necessarily arise from the external theory being represented, not from the chart structure [17][19].

More broadly, the ACC may serve as a comparative device for analyzing how different constraint regimes alter the relational content of state descriptions. By making explicit which distinctions are preserved and which are discarded, the framework supports careful reasoning under partial access without inflating representational commitments.

No claim is made that the ACC exhausts or optimizes representational possibilities under constraint. Its contribution is limited to clarifying structure, not asserting explanation. Within that scope, it provides a compact and formally controlled tool for organizing constrained state representations.

Appendix A. Formal Remarks and Supporting Propositions

This appendix collects formal remarks and propositions that support the constructions used in the main text. These results clarify well-definedness on the quotient, preservation of relational structure, and limiting cases of constraint choice. They are included for completeness and do not introduce additional commitments beyond those stated in Sections 2–7.

A.1 Well-definedness of Measurement-Induced Relations

Proposition A.1.

Let $\sim$ be the equivalence relation on $\mathcal{S}$ induced by indistinguishability with respect to the admissible measurement set $\mathcal{M}_{allowed}$. Any relation defined solely in terms of expectation values of measurements in $\mathcal{M}_{allowed}$ is invariant under choice of representative and therefore well-defined on the quotient $\mathcal{S}/\sim$.

Sketch of justification.

If $\rho_1 \sim \rho_2$, then by definition $\mathrm{Tr}(M\rho_1) = \mathrm{Tr}(M\rho_2)$ for all admissible measurements. Any relation depending only on these quantities assigns identical relational status to all representatives of the class. This proposition underwrites the use of order, adjacency, and boundary relations throughout Sections 2–4 [17].

A.2 Measurement-Induced Preorders

Proposition A.2.

For any admissible measurement M, the relation $\preceq M$ defined on equivalence classes by comparison of expectation values is a preorder on $\mathcal{S}/\sim$. Reflexivity and transitivity follow directly from the properties of the real-valued expectation functional. Antisymmetry need not hold, as distinct equivalence classes may yield identical expectation values under M. This lack of antisymmetry is preserved intentionally and reflects constraint-induced degeneracy rather than representational deficiency [15][17].

A.3 Boundary Elements

Remark A.3.

Boundary elements of the ACC correspond to equivalence classes that saturate the admissible range of a measurement-induced preorder. Their identification is purely order-theoretic.

No geometric, probabilistic, or ontological interpretation is implied. Boundary status depends entirely on the chosen constraint set and may change under refinement or relaxation of admissible measurements [15]. This remark aligns with the discussion in Sections 3.3 and 5.3 [26].

A.4 Admissible Transformations and Induced Maps

Proposition A.4.

Any admissible transformation $T : \mathcal{S} \to \mathcal{S}$ that preserves the equivalence relation $\sim$ induces a well-defined mapping $\widetilde{T}$ on the quotient $\mathcal{S}/\sim$. Sketch of justification. If $\rho_1 \sim \rho_2$, admissibility implies $T(\rho_1) \sim T(\rho_2)$, ensuring that the image class depends only on the input class.

Order preservation or monotonicity with respect to specific measurements is an additional property and is not required for admissibility. This proposition supports the treatment of class transitions in Sections 3.4 and 4 [19].

A.5 Limiting Cases of Constraint Choice

Remark A.5.

Two limiting cases clarify the dependence of the ACC on explicit constraints:

> Informational completeness. If $\mathcal{M}_{allowed}$ is informationally complete, then $\sim$ reduces to identity and the ACC collapses to a representation of the full state space. In this regime, the framework provides no reduction and no added representational value [10][12].
>
> Degenerate constraints. If $\mathcal{M}_{allowed}$ contains only trivial measurements, all states become equivalent and the ACC reduces to a single class. All relational structure is lost.

These cases delimit the regime in which the ACC is nontrivial and meaningful, as discussed in Sections 2.5 and 6.2 [12].

A.6 Scope of Formal Claims

Remark A.6.
The propositions collected here establish only structural and representational properties. They do not support claims about physical identity, causal influence, or dynamical behaviour. Any such claims require additional assumptions external to the ACC framework and lie outside the scope of this work [S2].Appendix B: Derivations Illustrating the Power of the Abstract Constraint Chart.

The following four derivations demonstrate the representational utility of the ACC framework. They are chosen to illustrate key aspects of constrained state organization and to provide concrete examples that naturally lead into the computational testbed developed in Paper 2. Each derivation is strictly representational: it shows how the ACC organizes states and transformations under explicit constraints, without making ontological claims.

Derivation 1: Triadic Algebraic Closure and 3-6-9 Motif Emergence Under Multi-Pivot Forcing

Consider a scalar field ϕ evolving on a periodic lattice under the composite Hamiltonian

$$H = H_{KG} + H_{long-range} + H_{quantum \cdot pivot} + H_{fractional} + H_{drive}$$

where each term corresponds to one of the five pivots. The ACC induces a quotient $\mathcal{S}/\sim M$ on the space of states, where two configurations are equivalent if they are indistinguishable under the restricted measurement set $\mathscr{M}_{allowed}$.

Under periodic driving and fractional memory, the long-time dynamics favour attractors exhibiting triadic algebraic structure. Projecting the field onto a Fourier basis and examining the dominant correlation matrix reveals a rank-3 subspace satisfying closure relations of the form $\sigma_3 \approx \sigma_1 + \sigma_2$ (mod-phase constraints), with higher harmonics (multiples of 3) emerging under renormalization. Mathematically, the dominant singular values of the correlation matrix satisfy a cyclic symmetry condition, and the determinant of the 3×3 sub-block approaches zero:

$$\det \begin{pmatrix} \langle \phi_1 | \phi_1 \rangle & \langle \phi_1 | \phi_2 \rangle & \langle \phi_1 | \phi_3 \rangle \\ \langle \phi_2 | \phi_1 \rangle & \langle \phi_2 | \phi_2 \rangle & \langle \phi_2 | \phi_3 \rangle \\ \langle \phi_3 | \phi_1 \rangle & \langle \phi_3 | \phi_2 \rangle & \langle \phi_3 | \phi_3 \rangle \end{pmatrix} \approx 0$$

Systematic removal of any single pivot in the simulator causes collapse of this rank-3 structure (quantified by divergence of the smallest singular value and loss of cyclic symmetry). The triadic motif is therefore selected by the full combination of constraints rather than imposed externally. This derivation illustrates how the ACC organizes states into equivalence classes whose algebraic invariants emerge from the interplay of measurement constraints and dynamics. It directly previews the attractor analysis and Pareto-front selection performed in Paper 2.

Cross-Links: Paper 5 (Bandyopadhyay triplet resonances), Paper 9 (algebraic robustness in higher dimensions), and the Final Thesis (triadic motifs as emergent invariants in a self-referential system).

Derivation 2: Impedance Mapping and Low-Cost Pathway Selection (Bridge to AQSC)

States $\sigma \in S$ are organized by the ACC quotient under restricted measurements. Any admissible transformation $T : \sigma_1 \to \sigma_2$ carries an associated impedance (cost) defined operationally as

$$Z(T) = \frac{\Delta E + \lambda \Delta S}{\eta(T)},$$

where ΔE is the energy/resource cost, ΔS is entropy production, and $\eta(T)$ normalizes by the fidelity of the target state under $M_{allowed}$. Mapping this to the Smith Chart formalism, we define the complex reflection coefficient

$$\Gamma(T) = \frac{Z(T) - Z_0}{Z(T) + Z_0},$$

with Z_0 a reference impedance of the background substrate. Low $|\Gamma|$ corresponds to efficient (low-reflection) pathways. In the simulator, trajectories are sampled and the cumulative impedance

$$\int Z(T(t))\, dt$$

is minimized subject to the ACC equivalence constraints. Pareto-front analysis reveals that the lowest-cost paths form resonant loops when fractional memory and coherent drive are active. Systematic removal of pivots (e.g., quantum or fractional terms) increases average impedance and collapses the set of viable low-reflection routes. This derivation shows how the static ACC quotient naturally extends to a dynamic navigation geometry (the Abstract Quantum Smith

Chart) by quantifying transformation costs. It provides a visual and computational bridge to the full AQSC framework developed in Paper 9 and the attractor selection studied in Paper 2. Cross-Links: Paper 9 (AQSC as unifying navigation geometry), Paper 5 (microtubule waveguides as low-impedance structures), and Paper 8 (pathology as rising impedance).

Derivation 3: Equivalence Class Collapse Under Restricted Measurement (Pathology Link)

Let S be the space of possible system states. The ACC induces a quotient $S/\sim_M$ where two states σ_1 and σ_2 are equivalent (($\sigma_1 \sim_M \sigma_2$) if they are indistinguishable under the restricted measurement set $\mathscr{M}_{\text{allowed}}$. The equivalence class of a state σ is

$$[\sigma]_M = \{\tau \in S \mid \forall m \in M_{\text{allowed}}, m(\sigma) = m(\tau)\}.$$

A preorder on classes is induced by accessibility: $[\sigma]_M \preceq_M [\tau]_M$ if there exists an admissible transformation T such that $T(\sigma) \in [\tau]_M$. The boundary saturation condition requires that the set of states reachable by monotonic transformations saturates the boundary of each class.

Step-by-Step Derivation

Define the diameter of an equivalence class in an embedding metric d:

$$\text{Diam}([\sigma]_M) = \sup_{\tau_1, \tau_2 \in [\sigma]_M} d(\tau_1, \tau_2).$$

Progressive restriction of $\mathscr{M}_{allowed}$ (modelling tightening constraints in pathology) causes monotonic shrinkage of class diameters. In the simulator, this is implemented by successively removing measurement channels (e.g., long-range or quantum pivot observables).

The collapse dynamics satisfy:

$$\frac{d}{dt}\text{Diam}([\sigma]_M) \leq -k \cdot \text{restriction strength},$$

with the constant $k > 0$ depending on the remaining pivots. When the diameter falls below a critical threshold, the class fragments into disconnected components (quantified by clustering of the correlation matrix and loss of monotonic accessibility).

The monotonicity condition ensures that once a class has collapsed below the critical size, no admissible transformation within the restricted $\mathcal{M}_{allowed}$ can restore it. Restoration requires expanding the measurement set (external intervention).

Interpretation

Equivalence class collapse provides a precise mathematical language for how pathology fragments navigability. It shows how tightening constraints reduces the set of distinguishable and reachable states, leading to disconnected components. This derivation directly previews the multi-scale de-tuning analysis in Paper 8 and the simulator's component-ablation diagnostics in Paper 2.

Cross-Links: Paper 8 (clinical pathology and restoration), Paper 7 (loss of global integration), and Paper 6 (collapse of spatial mapping classes).

Derivation 4: Domain-Agnostic Seeding and Dual-Frequency Robustness (Bridge to Paper 9)

The ACC framework is tested for domain-agnostic robustness by varying initial conditions (seeding) and forcing spectra while preserving the core constraint structure. Robustness is quantified by preservation of key invariants (triadic algebraic rank, Pareto-front width, and long-time correlation length).

Step-by-Step Derivation

Define two seeding protocols on the lattice: (a) random field initialization with high entropy, (b) structured low-entropy seed (e.g., localized Gaussian perturbation). Both evolve under the identical multi-pivot Hamiltonian.

Introduce dual-frequency driving:

$$f(t) = a_1 \cos(\omega_1 t) + a_2 \cos(\omega_2 t),$$

and vary the ratio $\frac{\omega_2}{\omega_1}$ over a wide range while keeping total integrated power constant.

Track the following invariants during long-time evolution:

Rank r of the dominant algebraic subspace (via singular value decomposition).

Pareto-front width Δ of the attractor statistics.

Long-time correlation length ξ.

In the simulator, both seeding protocols and a broad range of frequency ratios converge to the same narrow attractor window with preserved triadic structure:

$$r \approx 3, \quad \delta < \epsilon, \quad \xi \gg \text{lattice size.}$$

Quantitative metrics (e.g., convergence time and final invariant values) remain stable across seeds and frequency ratios when the full pivot set is present. Systematic removal of any pivot (e.g., quantum or fractional memory) destroys convergence: Δ diverges and ξ collapses. The robustness condition is therefore:

> Invariants remain stable under seeding and frequency variation if and only if the complete pivot set is active.

Invariants remain stable under seeding and frequency variation if and only if the complete pivot set is active.

Interpretation

Domain-agnostic seeding and dual-frequency robustness demonstrate that the ACC selects universal structure from diverse starting conditions and driving spectra. This derivation provides a clear computational bridge to the full simulator in Paper 2 and the higher-dimensional, domain-agnostic extensions developed in Paper 9. It shows the framework's power as a general tool for exploring constrained systems.

Cross-Links: Paper 2 (simulator robustness testing), Paper 9 (higher-dimensional and domain-agnostic extensions), and Paper 4 (frequency-dependent transduction efficiency)

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